CDA 106
Numerical Methods for Data Science

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LECTURE-1/ January 20, 2026

About me

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Disease modeling

- Dynamics of diseases
- Differential Eqns
- TB/HIV/Covid-19
- - -

About this course "CDA 106"

Numerical Methods

✓ Numerical Linear Algebra

✓ $AX = b \Rightarrow X = A^{-1}b$

$A = BC$

$BCX = b$

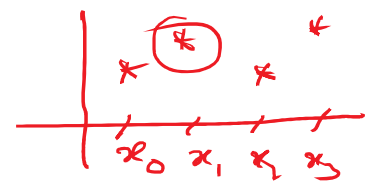
$AX = \lambda X$

Huge

Numerical Differentiation & Integration

• $y = y(x) \Rightarrow \frac{dy}{dx} = ?$

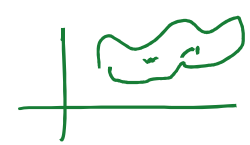
Int.



Diff.

Numerical Optimization

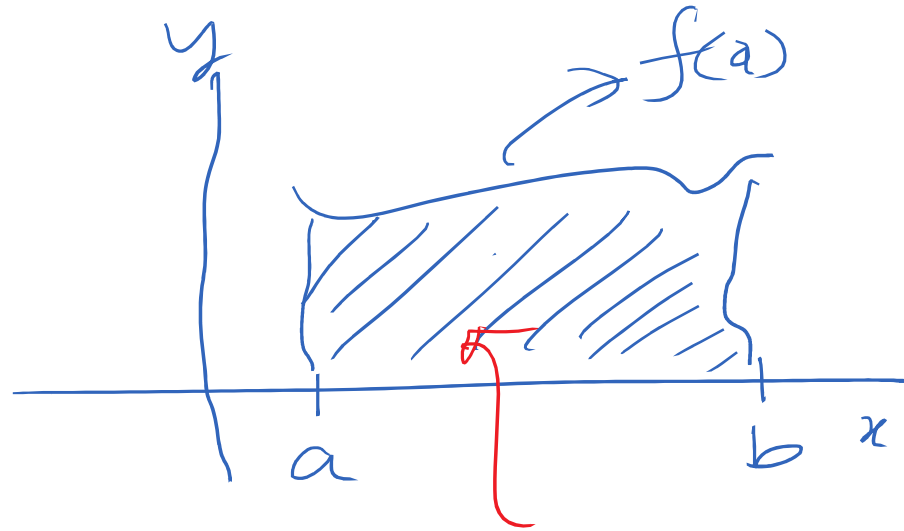
Max/Min



Numerical Integration

$$I = \int_a^b f(x)\,dx$$

= Area under the curve



Area : ?

$$F'(x) = f(x)$$

$$I = \int_a^b F'(x)\,dx = F(b) - F(a)$$

Consider

$$\int_0^1 e^{-x^2}\,dx = ?$$

$$\int_0^1 \frac{e^x}{x}\,dx \qquad \int_0^1 \frac{\log x}{x^2}\,dx$$

To solve such integrals we need Numerical technique!

Instead of given a function $y = f(x)$

x :	x_0	x_1	x_2	x_3	$\cdots$	x_n
y :	y_0	y_1	y_2	y_3		y_n

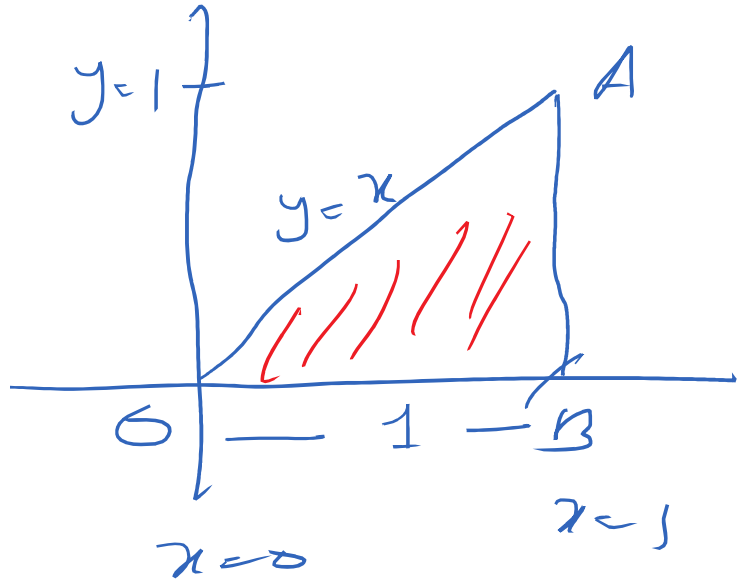
<u>Area</u> = ?

<u>Numerical Int</u>

Area of $\triangle OAB$

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$



$$\triangle = \int_0^1 x\,dx = \frac{x^2}{2} = \frac{1}{2}$$

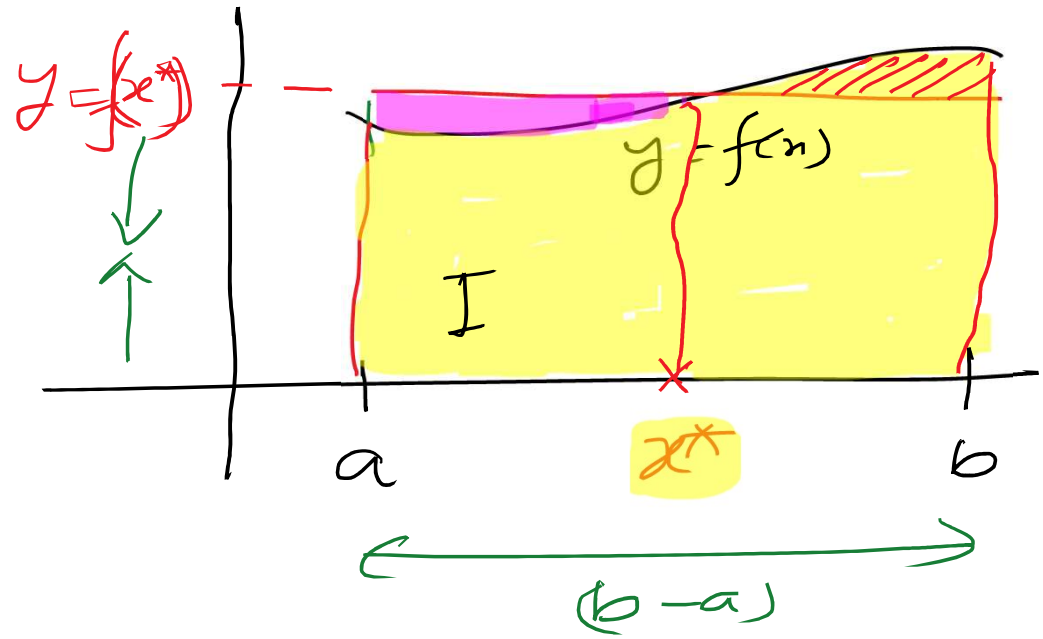
1) Rectangular Method

$$I = \int_a^b f(x)\,dx$$

$$\simeq (b-a) \times f(x^*)$$

$$\simeq A$$

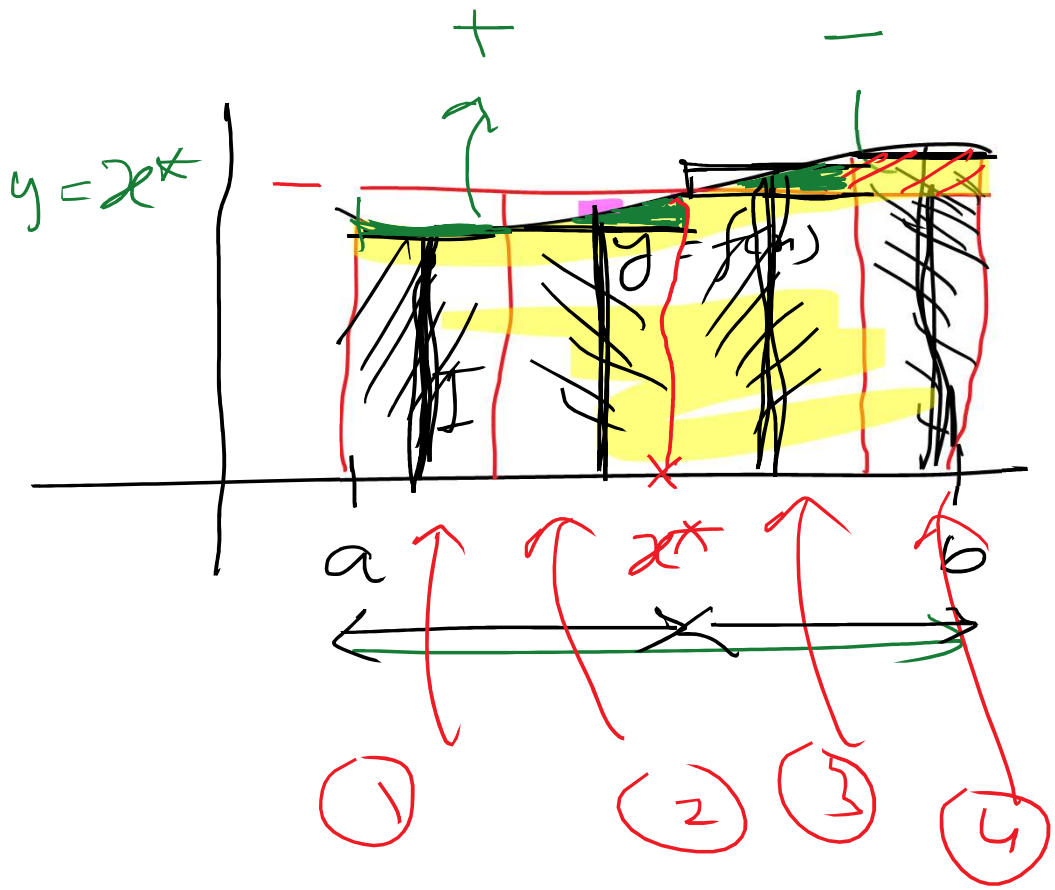
$$= (b-a)\, f\left(\frac{a+b}{2}\right)$$



Area $I = A$ with some error

Area of sum of rectangle

$\longrightarrow$ Area under curve



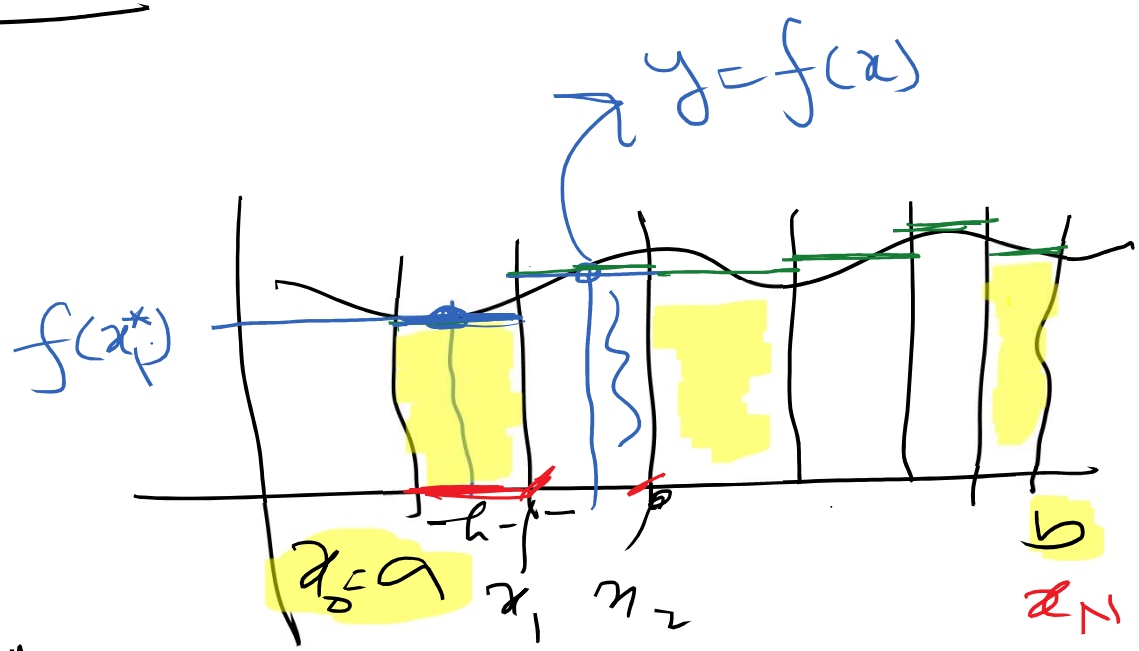
Rectangular Method

① → Partition $[a,b]$ in N parts

$$\frac{b-a}{N} = h$$

$$x_j = a + jh, \quad 0 \le j \le N$$

$$\{\underset{=a}{x_0}, x_1, x_2, \dots, \underset{=b}{x_N}\}$$



$$x_1^* = \frac{x_0 + x_1}{2}$$

$$x_j^* = \frac{x_{j-1} + x_j}{2}$$ is midpoint $[x_{j-1}, x_j]$

$I =$ Sum of Area of all the rectangles

$$= h \times f(x_1^*) + h \times f(x_2^*) + \cdots + h f(x_N^*)$$

$$= h \left[f(x^*) + f(x_2^*) + \cdots + f(x_N^*) \right]$$

$$\underline{\qquad\qquad} \times \underline{\qquad\qquad\qquad\qquad}$$

Ex To find Area of ΔOAB

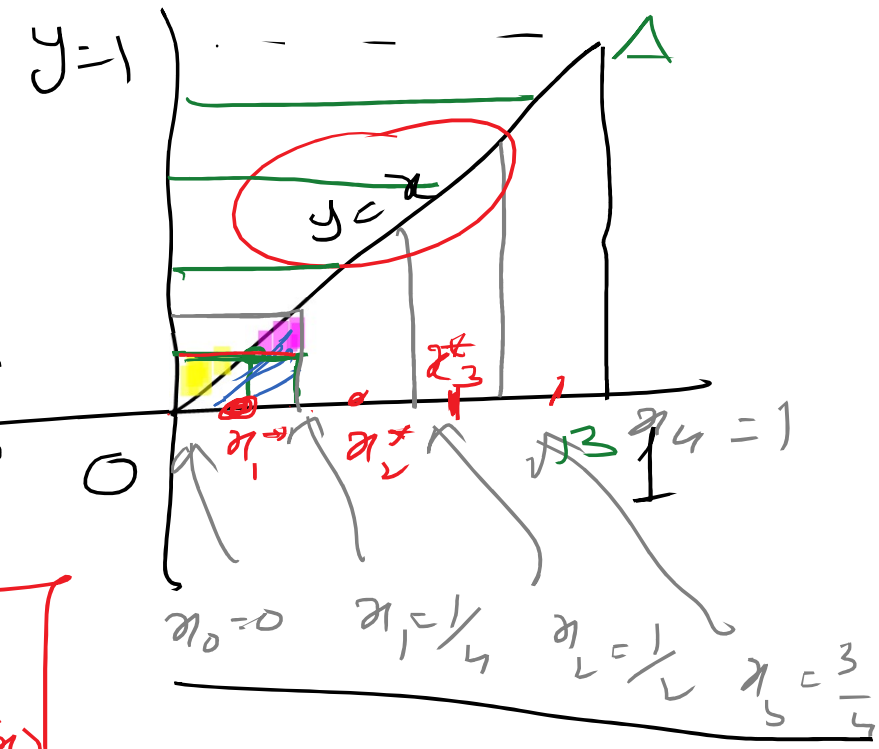
$$h = \frac{1-0}{4} \quad (\text{Choose } N = 4)$$

$$= 1/4$$

Check

$$x_j = 0 + j\,h$$

$$0 \le j \le 4$$



$$x_1^* = \frac{1}{8} \; ; \; x_2^* = \frac{3}{8}$$

$$x_3^* = \frac{5}{8} \qquad x_4^* = \frac{7}{8}$$

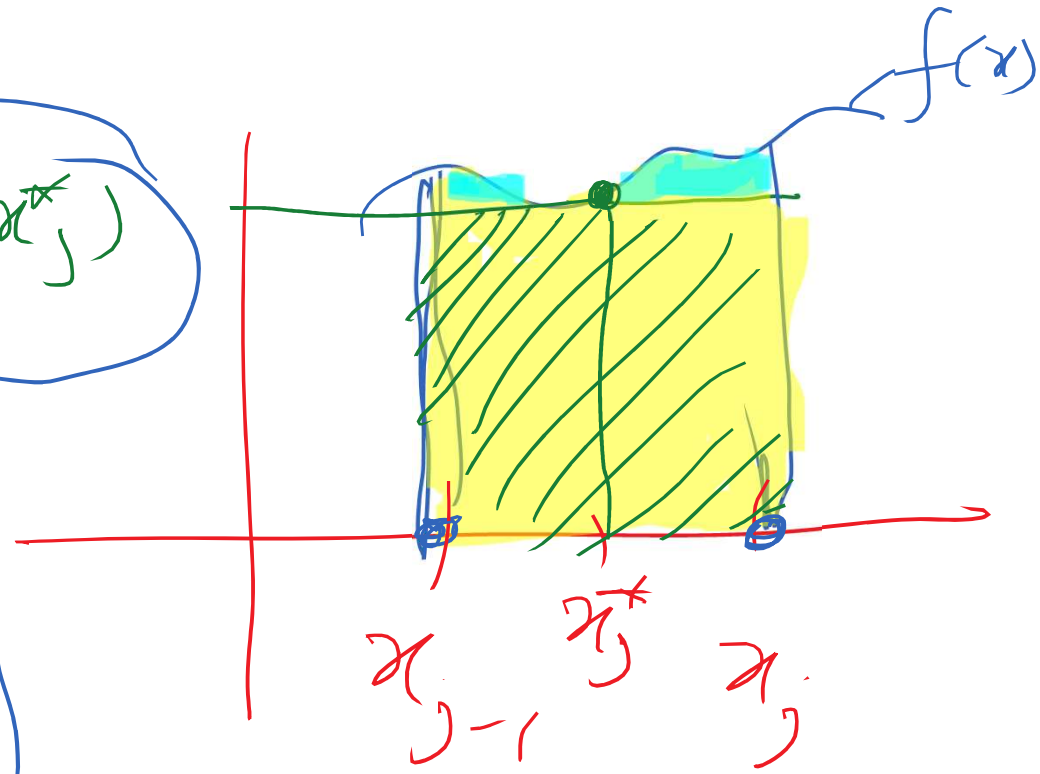
$$\boxed{y = x = f(x)}$$

$$I = h\left[f(x_1^*) + f(x_2^*) + f(x_3^*) + f(x_4^*)\right] = \frac{1}{4}\left[x_1^* + x_2^* + x_3^* + x_4^*\right] = \frac{1}{4} \times \frac{16}{8}$$

$$= 1/2$$

$$f(x) = f(x_j^*)$$

$f(x_j^*)$

= constant value

on $[x_{j-1}, x_j]$